Reg. No. | | | | | | | | | | |

[18MSCH3S31]

THIRD SEMESTER UNIVERSITY EXAMINATION, DECEMBER 2021

APPLIED CHEMISTRY

Time : 03 Hours **Maximum Marks:70**

Instructions: **1. Question 1 is compulsory, carries 10 marks.**

2. Questions 2 to 7 carries 15 marks each, answer any FOUR.

1. Answer any FIVE of the following questions. **5 x 2 = 10**

a. How nano size of the catalyst affects the chemical reaction?

b. What are the factors affecting the resolution of a microscope?

c. Explain anionic polymerization with examples.

d. What are chromophores? Give two examples.

e. Define surfactants. Explain their classification.

f. Write the Hammett and Bronsted equation for acid-base catalysis.

2. a. Describe different procedures employed for the synthesis of quantum dots. Explain their applications.

b. Explain the chemical vapour deposition process for graphene synthesis with diagram.

c. What are semiconductor quantum dots? Explain their optical properties.

(5+5+5)

3. a. Explain the working principle of scanning electron microscope with a schematic diagram.

b. Briefly explain the principle of electron energy-loss spectroscopy. **(10+5)**

4. a. Explain the free radical mechanism involved in styrene polymerization.

b. Write notes on Zeigler Natta catalyst and its role in polymerization.

c. What are thermosetting and thermoplastic polymers? Explain with suitable examples.

(5+5+5)

5. a. Define binders. Write a note on alkyd, epoxy and polyurethane resins.

b. Explain the synthesis and applications of crystal violet. **(10+5)**

6. a. Describe the Du Nouy Ring method for surface tension measurements.

b. Explain the process of Aerogel preparation with an example.

c. Explain the methods used for colloidal purification. **(5+5+5)**

7. a. Illustrate the mechanism and kinetics of acid-base catalysis with examples.

b. Describe autocatalysis and oscillatory reactions. **(10+5)**

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